Countdown Clock

Working with electricity and electronics can be extremely dangerous and may result in serious injury, death, electric shock, fire, or property damage. These instructions are provided for informational and educational purposes only.

- Always consult and follow local electrical codes, regulations, and manufacturer guidelines.
- High voltage and mains electricity should only be handled by qualified, licensed professionals.
- If you lack experience or are unsure, do not attempt any project—seek help from a certified electrician.
- Some diagrams or photos may omit safety equipment for clarity but do not imply it is unnecessary.

Use of these instructions is entirely at your own risk. The provider disclaims all liability for any damage, injury, loss, or expense arising from following or misapplying this information. Your safety is your sole responsibility.

For the full build video see https://youtu.be/xp5f11BEO7U

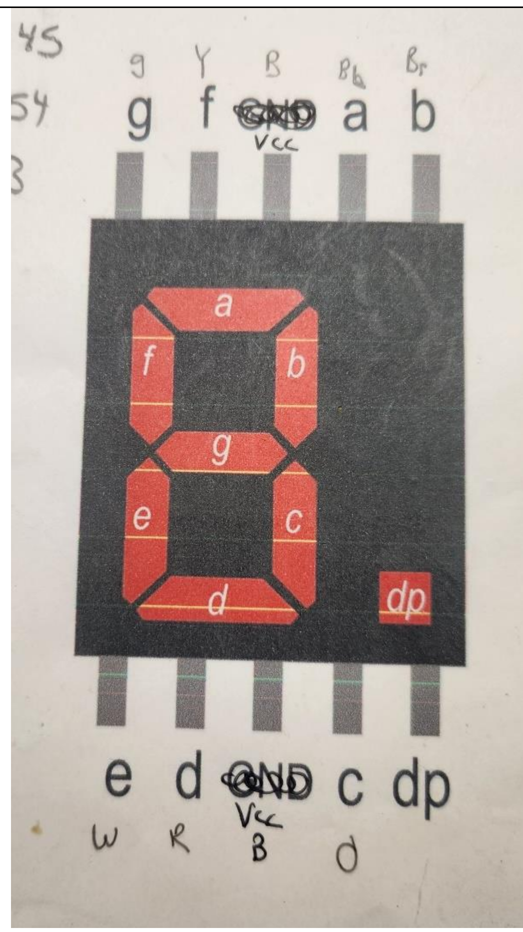
Parts: I used AliExpress for parts as they were just so much cheaper. A generic ESP32C3 will be fine for this build also.

- 74HC595 Expansion Module https://www.aliexpress.us/item/3256807085990530.html
- ESP32-C6 or C3 supermini https://amzn.to/3ZNxFkJ
- Liquid Electrical Tape https://amzn.to/4bvsXzn
- Tin-Plated Electronic Wires Breadboard PCB Jumper Wire 26AWG 6Color https://www.aliexpress.us/item/3256809442172424.html
- Two additional colors. I used servo wire like this with brown and orange. https://www.aliexpress.us/item/3256808060418939.html
- Ultra-long Type-C charging cable https://www.aliexpress.us/item/3256807723723506.html

- 7 Segment Red 5161BS
  https://www.aliexpress.us/item/3256811432929056.html
- 21 each, 1/8-Watt Metal Film Resistors 1K Ohm
- Heat shrinkable tube 2mm dia
  https://www.aliexpress.us/item/3256809147827721.html
- **Optional**
  5V Led Strip Light SMD2835 120 Leds/m
  https://www.aliexpress.us/item/3256804598752019.html

Software/ Schematics and printable instructions:

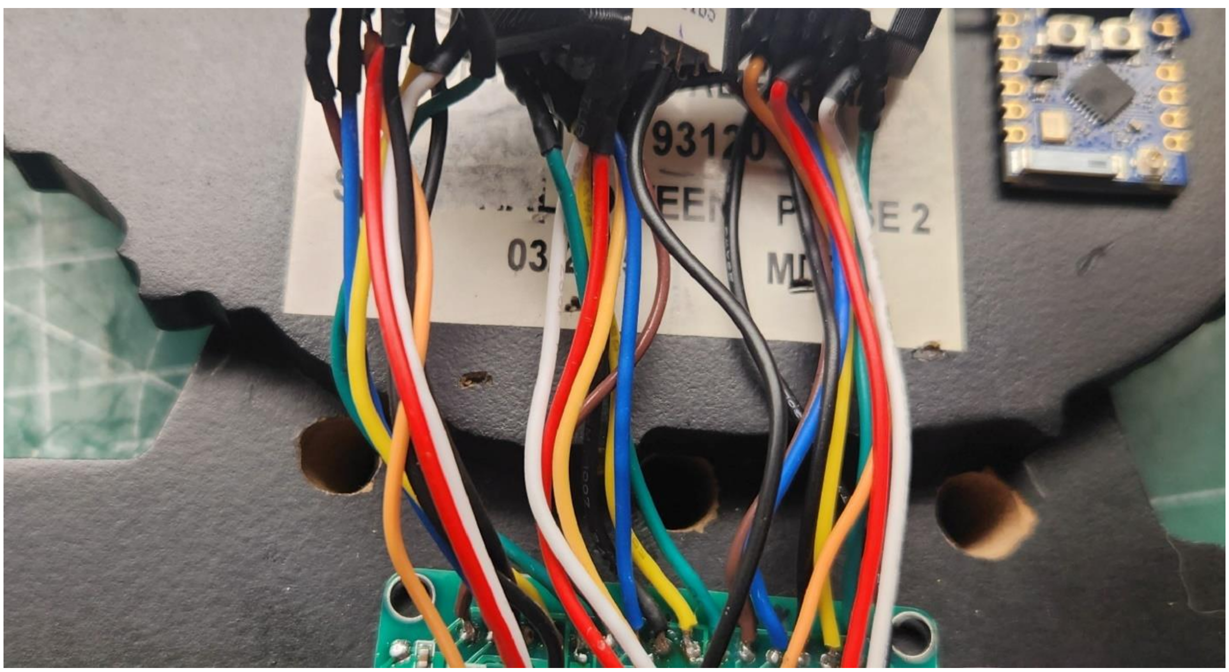
Module and display build:

You will start building the wiring from the displays. I printed out this image and assigned colors to each segment. Consistency and testing are key. Make sure to use a resistor when testing so you don't release the magic smoke. Every segment will need a resistor. The decimal point(dp) is not used and VCC do not need a resistor either. I tested and found the 1K resistor is just right for this application for me. You can use a lower value if you want brighter segments. Common static driven 7-segment displays have 470-ohm resistors. I just found this too offensive indoors. You will need to trim a bit of the wires from the displays to ensure they will fit in the cutout on the clock. After soldering I tested each segment before shrink wrapping. Once complete the wires will get folded in towards the center to fit in the cutout.	

In my build I used Common Anode. Although I used black wires to attach the VCC pins, these wires are connected to VCC(5v). When testing a 7-segment common anode display, you connect your red(positive) wire of the power supply to VCC of the display, pin 3 or 8, and the black (negative) of the power supply **via a resistor**, to an individual segment, pins a-g and dp. This is a big area of confusion. Common Anode displays are the most common used.

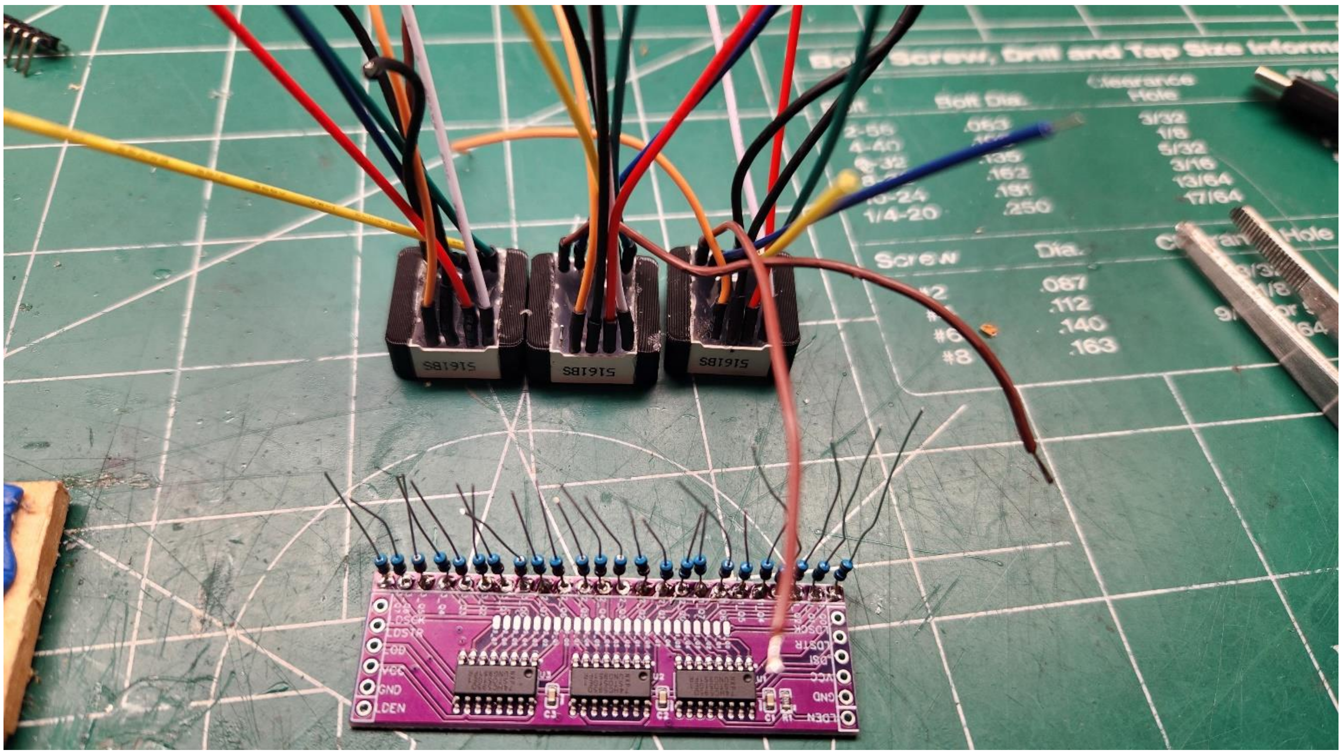
I used black wires for the VCC's. You don't really need both as they are connected internally in the display but I used two as a safety factor and so I could cascade the connections instead of a central buss bar.

Drilling access for the wiring into the clock is something you want to take your time with. If possible, avoid hitting the clock face. If you do it's not a big deal as you won't see it from the front. Drill from the front, inside the cutout to the back. I used ¼" drill bit and angled the holes to keep away from the clock face wood.

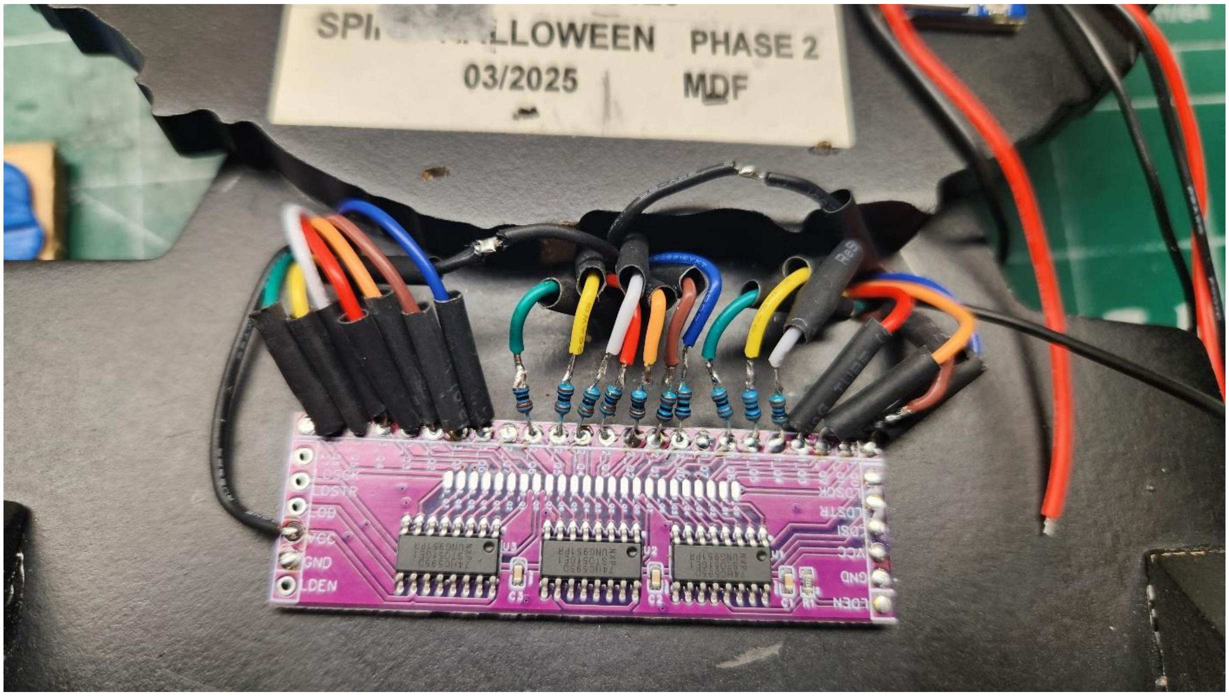


On the module you will need to solder your resistors to each output connection. With this application you will not need the decimal point so Q7 connections on each display will be empty. Leave a little wire of the resistor above the module so you have some flexibility.

After the displays are checked you can glue the 3D printed segment fillers to each side. It's a tight perfect fit so just a bit of glue firmly pressed. I used "Goop" but anything that bonds to plastic will work.

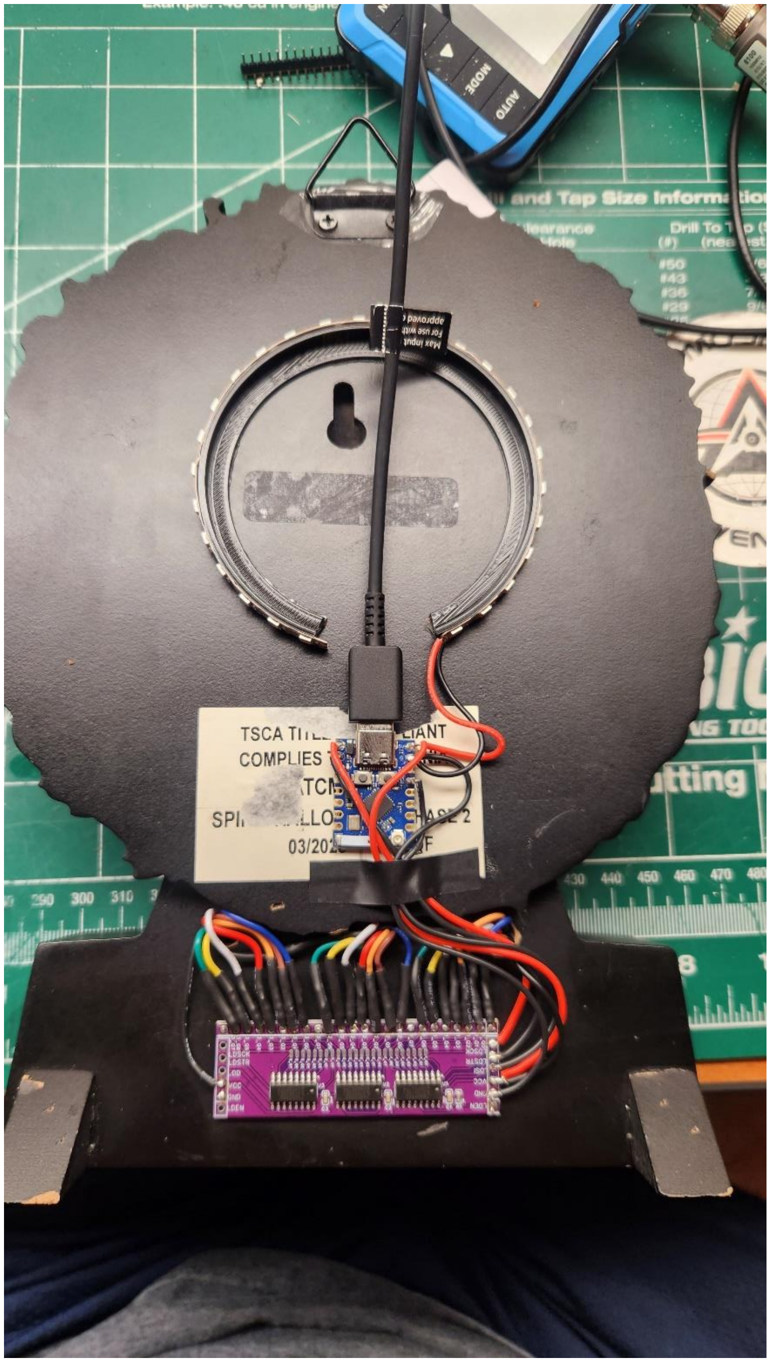


Segments are connected in order a, b, c, d, etc to the module outputs Q0, Q1, Q2, Q3 etc. There is no connection at 1_Q7, 2_Q7 and 3_Q7 as these would be used for the decimal point which is unused in this application.



Components were attached to the clock with double sided 3M VHB tape. Make all connections, test then come back and heat the heat shrink tube. This is very tedious. I just slid all the tubes over to ensure no shorting then when all was working came back and shrunk them down.

I wanted a little more effect to this build so I added some purple LEDs to cast a glowing effect. I tried a few different types before deciding on these. They are very tightly spaced. I also designed and printed a mount ring for them. Centered on the back it provides a nice purple glow. You could also use addressable LEDs and adapt the code to drive them via the app and change colors. I just wanted purple and felt not need for the extra addressable bling. I would have loved to use purple 7-segment displays but they could not be found.



I positioned the ring in the center of the USB cord so the light would not cast oddly due to the cord.
With the LEDs on this project is drawing 400ma so can be powered by the smallest USB power brick.

Programing and connecting to ESP32

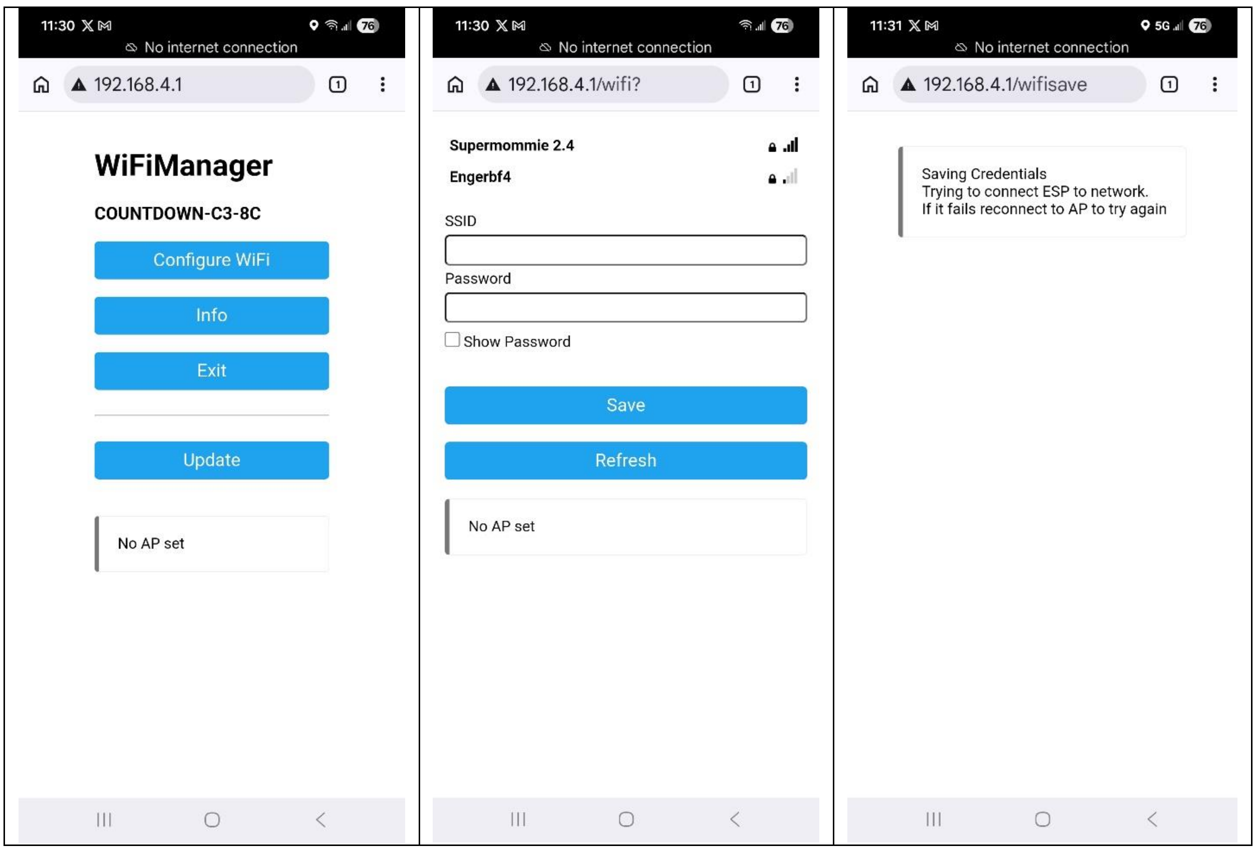
Upload the sketch "Countdown_Clock.ino" to your ESP32.

Initial setup:

When device is powered up, connect your phone to "Countdown-C3" Wi-Fi

The initial password is "countdown123"

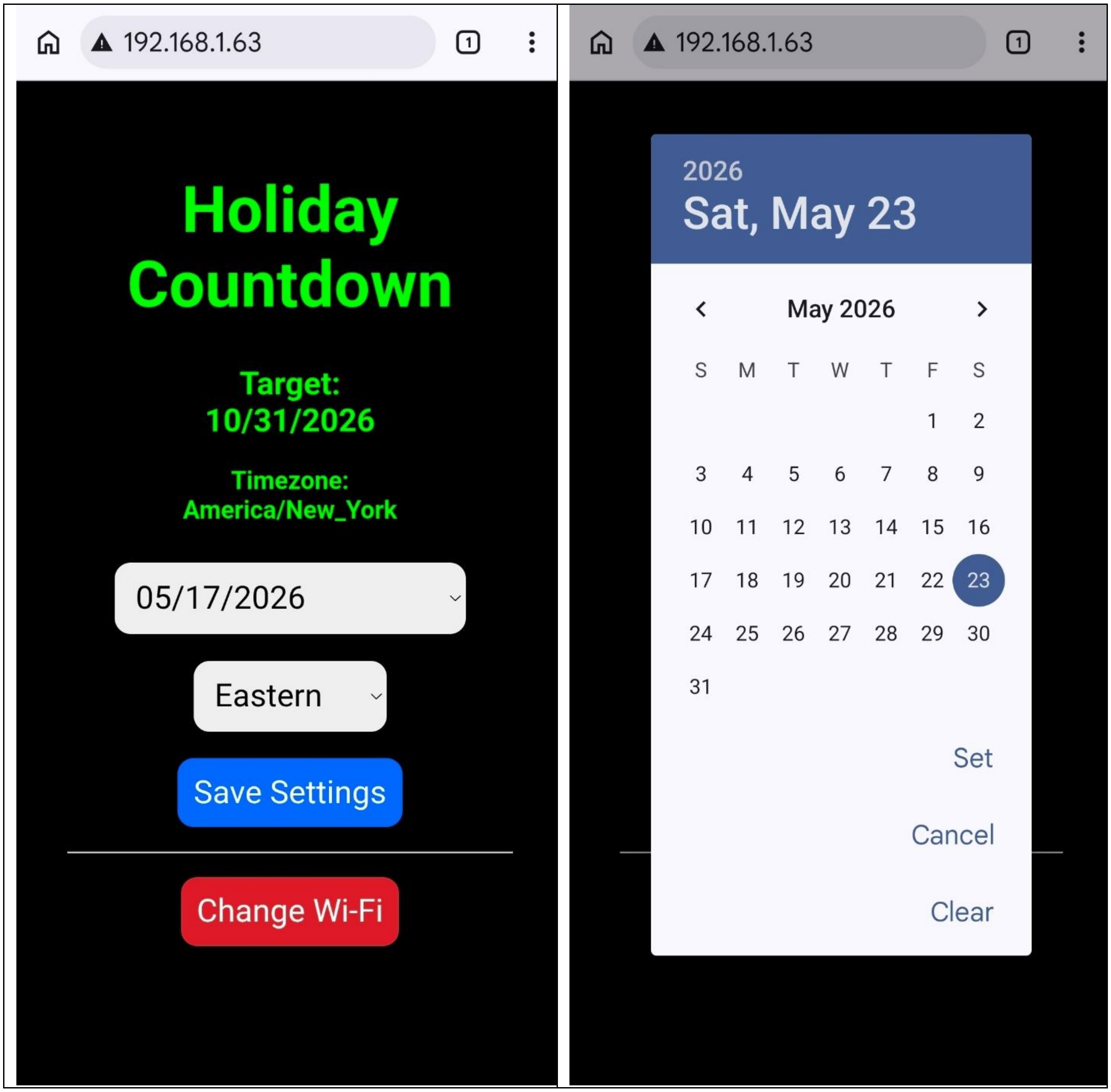
Now, go to 192.168.4.1 on your phone browser and enter your home Wi-Fi settings.



Click on your home Wi-Fi and it will populate in the text box. Then enter your password and the device will connect to your home network and your router will assign an address for this ESP32. You will need to log onto your router and find the unique address.

I was using an ESP32c3 and the name of the connected device was "esp32c3-xxxxxx". You should see an address like "192.168.1.63" (everyone's will be unique.)

Once connected to your home Wi-Fi, navigate your phone to that address in your web browser. You should see the webpage for configuring the target date, as well as time zone and a reset for the Wi-Fi network.



On my device I printed out a cheat sheet to attach for ease of configuring. Just change the custom address to what yours is.

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Initial setup:

Connect to “Countdown-C3” WIFI

password is “countdown123”

Go to 192.168.4.1

Set up home Wi-Fi settings

An address will be created on the router for this device

Currently set to _______________